

Assignment 2 Case Study

Stage 2 Tutorial From Problems to Requirements

Week 5 | 60 minutes

Learning goals

- Analyse stakeholders.
- Distinguish functional and non-functional requirements.
- Recognise ambiguity and unsupported requirements.
- Define scope.
- Develop user stories and acceptance criteria.
- Critique AI-generated requirements.

Activity 1 - Stakeholder Map

Stakeholder	Need	Potential conflict
Patients	Easy booking and cancelling, correct appointment details	Want flexible cancelling, while management may want to limit late cancellations
Practitioners (GPs)	See their schedule and availability; access patient details	Want control of their time, while reception wants to fit in as many bookings as possible
Reception/admin staff	Fast patient search; reliable booking and cancelling	Privacy and validation rules could slow their work
Clinic management	Basic reports, fewer errors, a simple and affordable system	Want a small scope, while staff and patients keep asking for extra features
Developers/maintainers	Clear, testable requirements	Vague or changing requests from other stakeholders

Activity 2 - Functional or Non-Functional?

- ☒ Functional ☐ Non-functional The system shall allow staff to cancel an appointment.
- ☐ Functional ☒ Non-functional The system should remain responsive for the course-scale dataset.
- ☒ Functional ☐ Non-functional The system shall retain cancelled appointments.
- ☐ Functional ☒ Non-functional Core business logic should be independently testable.
- ☒ Functional ☐ Non-functional The system shall search for a patient by ID.

Activity 3 - Repair Ambiguous Requirements

The system should be easy to use.

Problem: "Easy" is subjective and can't be measured. Clarification question: Which users, and what should a new receptionist be able to do without training, in how many steps?

Patient search should be fast.

Problem: No time limit or number of records is given. Clarification question: How many seconds is acceptable, with how many patient records?

The system should securely manage data.

Problem: Doesn't say which data, who has access or what protections apply. Clarification question: Which data is sensitive, who may see it, and what privacy rules apply?

Appointments should normally be easy to cancel.

Problem: "Normally" hides the exceptions, and "easy" is vague. Clarification question: When can't an appointment be cancelled (e.g. short notice), and who can cancel?

Activity 4 - AI Requirements Audit

Classify each suggestion: Confirmed / Assumption requiring validation / Unsupported / Out of scope.

AI suggestion	Classification	Evidence / reason
Patients receive SMS reminders.	Unsupported	The client never asked for reminders, and they would need phone numbers and extra cost.
Facial recognition login.	Unsupported	No evidence, and it raises privacy concerns for a small clinic.
Receptionists create appointments.	Confirmed	Staff booking is the core need, and duplicate bookings are a current problem.
Online payment.	Out of scope	Not requested, and beyond the first version.
Practitioners view schedules.	Confirmed	Limited visibility of availability is a stated problem.
AI recommends treatments.	Out of scope	It is a clinical decision, and the client does not want a hospital-style system.
Cancelled appointments remain in history.	Assumption requiring validation	No reliable history is a stated problem, but the retention rule has not been confirmed.

Exit question

Why is 'AI suggested it' not sufficient evidence for a requirement?

"AI suggested it" is not evidence, because AI can produce plausible features that nobody asked for and it does not know the clinic's real needs. A requirement needs support from the client, the case study or the stakeholders, and must be checked against them before it goes into the system.